
accelerate
collaborate
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PalmSource
Conference 2000



Platform Horizontal Tools

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Agenda

- Horizontal tools for Palm OS® application development
  - Application Development Tools
  - Conduit Development Tools
  - Web Clipping Development Tools

Application Development Tools

- Languages available
  - C/C++
    - CodeWarrior, Prc-Tools, Falch.net, VFDIDE, etc.
  - Basic Like
    - Satellite forms, Pendragon forms, NS Basic, etc
  - Java
    - IBM VisualAge Micro Edition
  - Others
    - Smalltalk, Lisp, Assembly, etc.

Application Development Tools

- Handheld Application Development Tools
  - CodeWarrior® for the Palm OS®
  - Prc-Tools
  - Falch.net DeveloperStudio
  - VFDIDE
  - Satellite Forms
  - Pendragon Forms
  - NS Basic
  - CASL
  - HanDBase
  - VisualAge Micro Editon for Palm OS (Java)

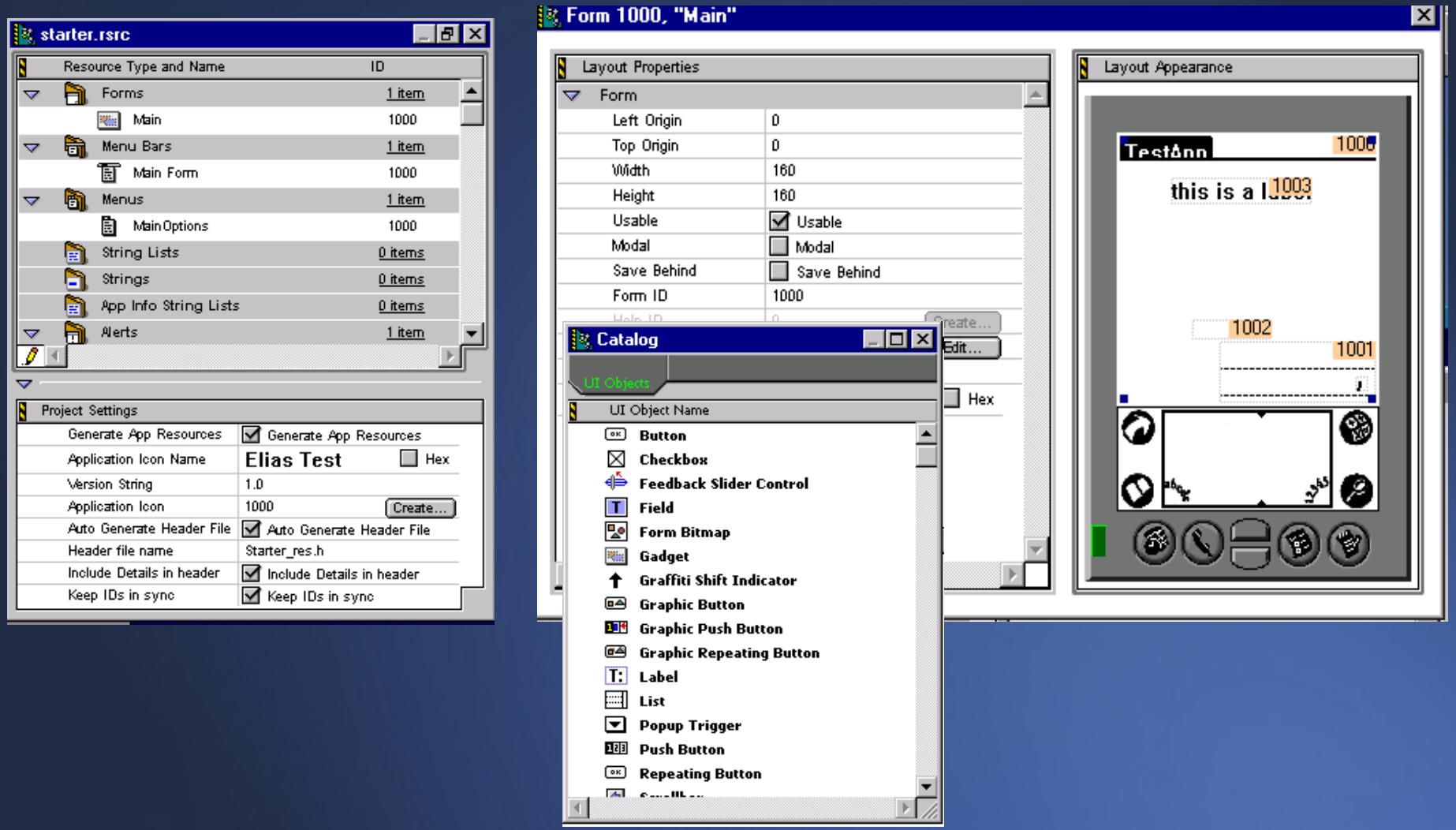
CodeWarrior for the Palm OS

- Integrated Development Environment C/C++
- Tool of choice among the Palm OS developers
- Creating, compiling, linking and debugging all within a single development environment
- Free for non-commercial use
- Cost $369.00

CodeWarrior® for the Palm OS®

- Features
  - C/C++ compiler
  - Palm OS Emulator (Windows and Mac)
  - Palm OS SDK
  - Constructor for visual design
  - Symbol SPT family Scanner Library
  - CodeWarrior IDE documentation and viewing applications
  - Palm OS documentation and tutorials
  - Project and version control

CodeWarrior for the Palm OS



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CodeWarrior for the Palm OS

- Why you want to use CodeWarrior
  - Palm OS API friendly
  - Build-in apps source code
  - Scalability
  - Free for non-commercial use

- http://www.palmos.com/dev/tech/tools/

Prc-Tools

- Complete tool chain for building Palm OS application in C or C++ (GCC)
  - GCC compiler, gdb debugger and PilPrc
  - Palm OS SDK and post-linker to produce .prc file
- Prc-Tools source code
- Support for Linux and Windows
- Free Free Free

- htpp://www.palmos.com/dev/tech/tools/gcc/

Falch.net DeveloperStudio

- Integrated Development Environment for C/C++
- Creates simple to most advanced apps
- Use Prc-Tools 2.0
- Free for non-commercial use
- Cost $149.00

Falch.net DeveloperStudio

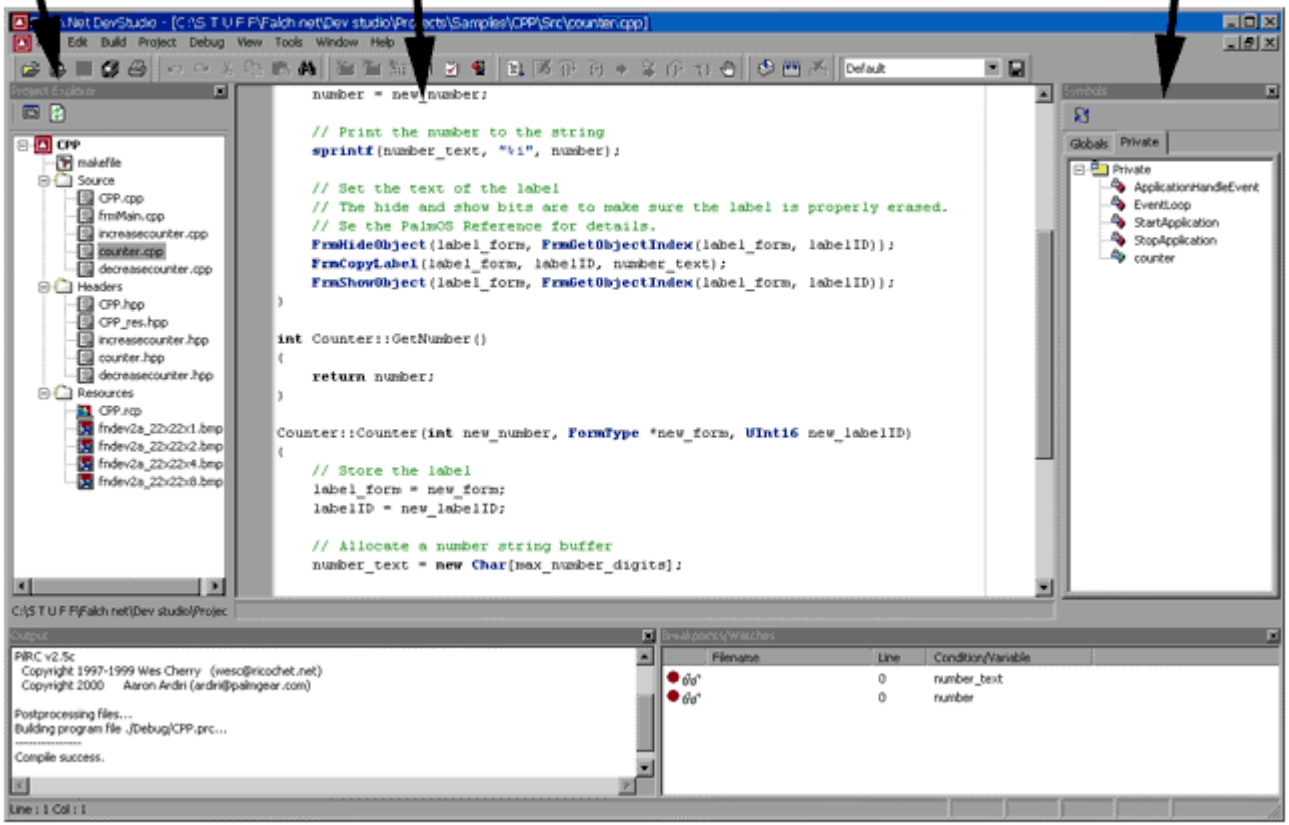
- Features
  - Easy to use text editor with full color syntax highlighting
  - Context sensitive online help
  - Code Wizard and Template support
  - Visual form designer and advanced resource editor
  - Integration with GCC, GDB and Palm™ Emulator (POSE)
  - Single click transfer of executables to the device
  - Open multiple projects in single workspace
  - Support for multiple breakpoints and watches

Falch.net DeveloperStudio

Project Explorer
With customizable folders

Source Editor
Includes syntax highlighting
of Palm OS API keywords

Symbols View
All symbols (functions and variables)
in the current project



Falch.net DeveloperStudio

- Why you want to use DeveloperStudio
  - Scalability from small to large applications
  - Full C/C++ support
  - Visual C++ like

- http://www.falch.net

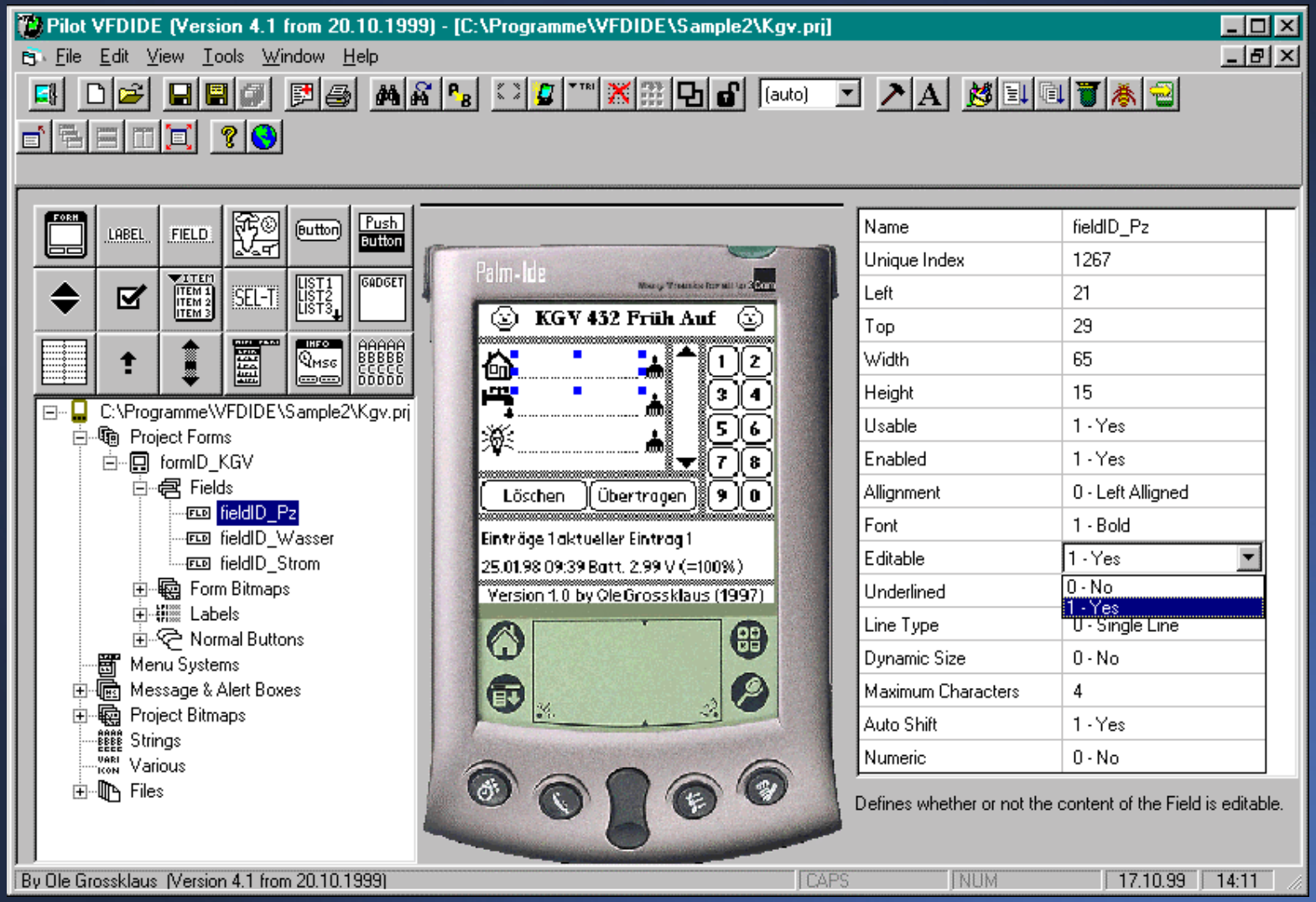
VFDIDE

- Visual Form Designer for C/C++
- Color Support
- Use Prc-Tool 2.0
- Cost $55

VFDIDE

- Features
  - True WYSIWYG Designer in gray as well as color
  - Integrated syntax highlighting, multi file source code editing
  - Line level debugging with break points, watch points, single step functions
  - Multi segment development
  - Free samples (Database management, communications, and games)

VFDIDE


Pilot VFDIDE (Version 4.1 from 20.10.1999) - [C:\Programme\VFDIDE\Sample2\Kgv.prj]
File Edit View Tools Window Help
(auto)
LABEL
FIELD
Button
Push Button
SEL-T
GADGET
C:\Programme\VFDIDE\Sample2\Kgv.prj
Project Forms
formID_KGV
Fields
fieldID_Pz
fieldID_Wasser
fieldID_Strom
Form Bitmaps
Labels
Normal Buttons
Menu Systems
Message & Alert Boxes
Project Bitmaps
Strings
Various
Files
Palm-Ide
KGV 432 Früh Auf
Löschen
Übertragen
Einträge 1 aktueller Eintrag 1
25.01.98 09:39 Batt. 2.99 V (=100%)
Version 1.0 by OleGrossklaus (1997)
Name fieldID_Pz
Unique Index 1267
Left 21
Top 29
Width 65
Height 15
Usable 1 - Yes
Enabled 1 - Yes
Allignment 0 - Left Alligned
Font 1 - Bold
Editable 1 - Yes
Underlined 0 - No
Line Type 1 - Yes
Dynamic Size 0 - No
Maximum Characters 4
Auto Shift 1 - Yes
Numeric 0 - No
Defines whether or not the content of the Field is editable.
By Ole Grossklaus (Version 4.1 from 20.10.1999)
CAPS
NUM
17.10.99
14:11

VFDIDE

- Why you want to use VFDIDE
  - Visual C++ programmer
  - Apps need to be scalable
  - Cost $55

- http://www.vfdide.com

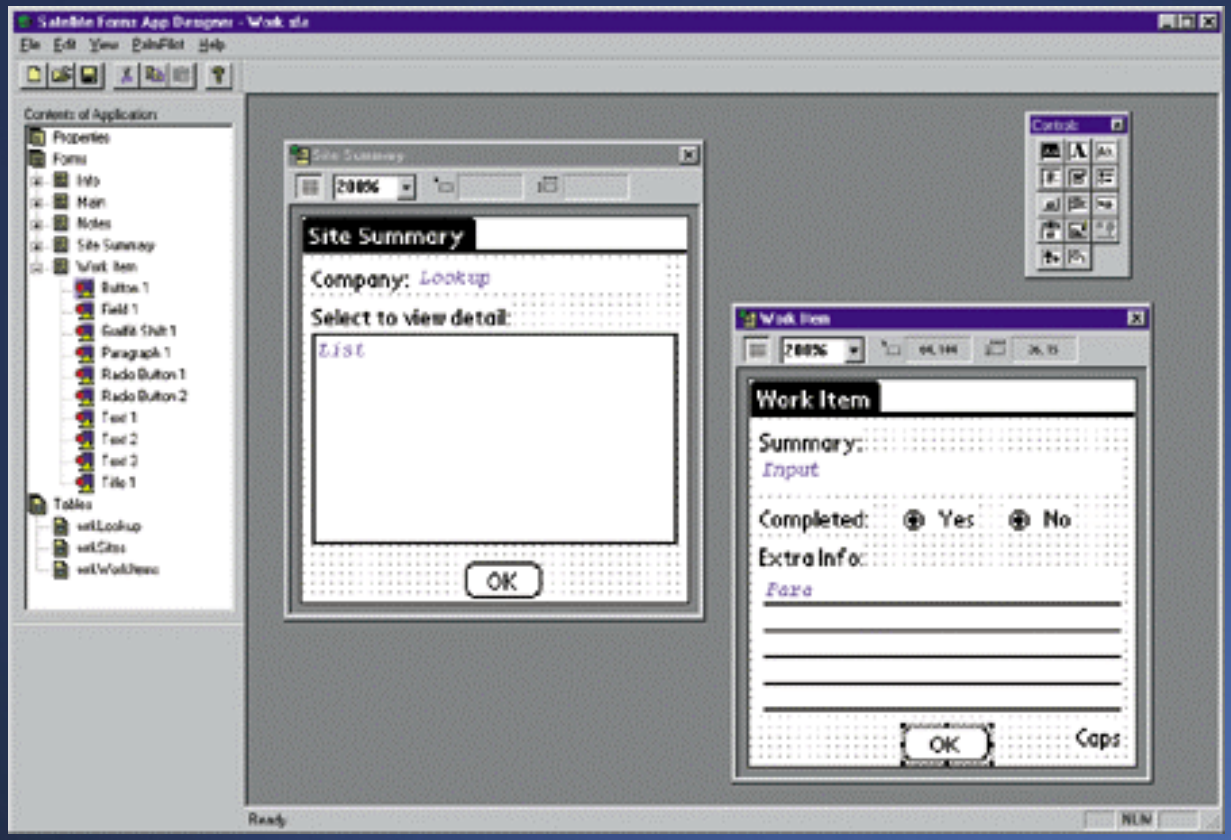
Satellite Forms

- Visual rapid application development
- Visual Basic like
- No programming
- Third party Plug-in support
- Cost $795

Satellite Forms

- Features
  - Create application with drag-and-drop
  - Multi-form applications with information sharing
  - ActiveX control for HotSync (Access, FoxPro, FileMaker, Dbase)
  - Custom forms designer
  - Scripting support to add more business rules, logic, calculation, and validation
  - Support for 3rd party controls and plug-ins

Satellite Forms



Satellite Forms

- Why you want to use Satellite Forms
  - Create “pen-friendly” forms with no computer experience or programming
  - Synchronize data at the press of a button
  - Need to develop quick database applications

- http://www.pumatech.com

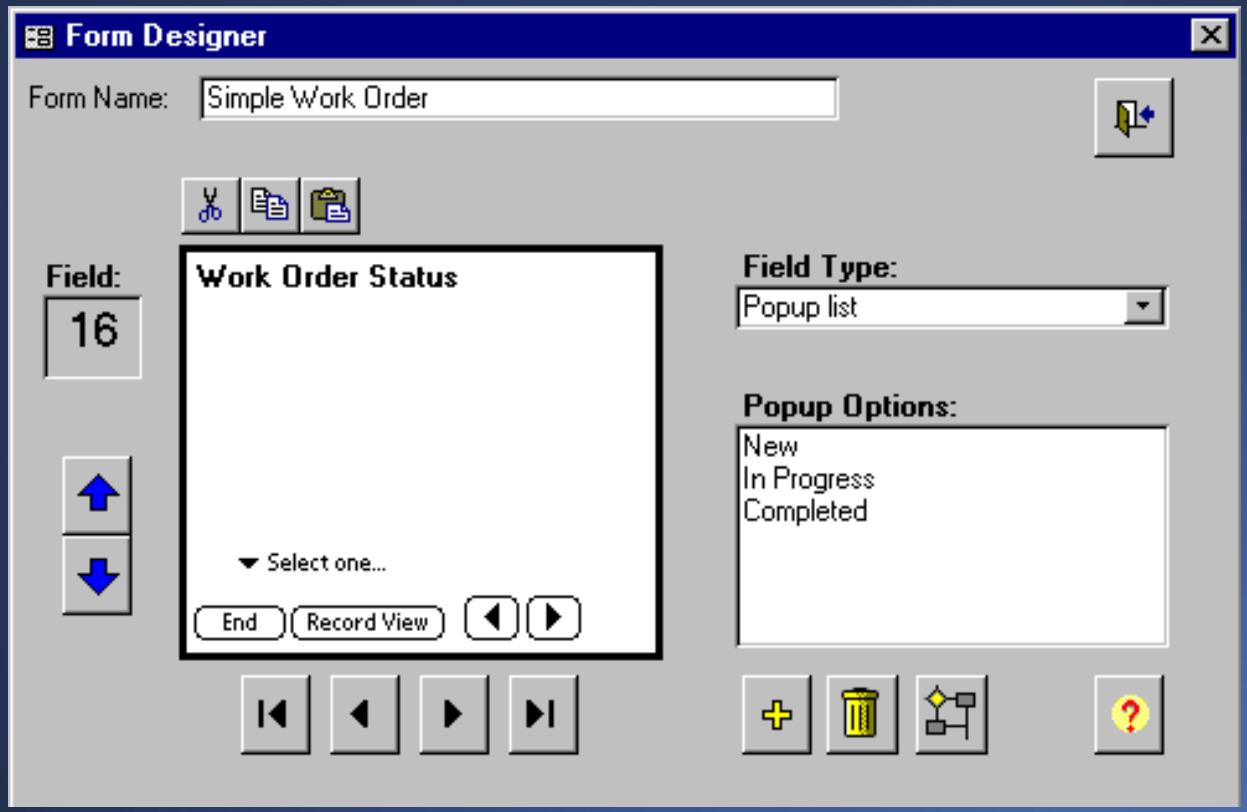
Pendragon Forms

- Rapid database form development
- Best for developing inventory control, patient tracking, work orders etc.
- Build a simple work order application in 15 minutes with no programming
- Cost $149.00

Pendragon Forms

- Features
  - Bi-directional synchronization with existing Access and ODBC databases
  - 21 different field types for data input and design
  - Map forms to external databases (Access and ODBC)
  - Generate forms based on external table design
  - Support for scripting language

Pendragon Forms



Pendragon Forms

- Why you want to use Pendragon Forms
  - Create data collection apps
  - Need to map forms with external tables
  - Cost $149.00

- http://www.pendragon-sofware.com

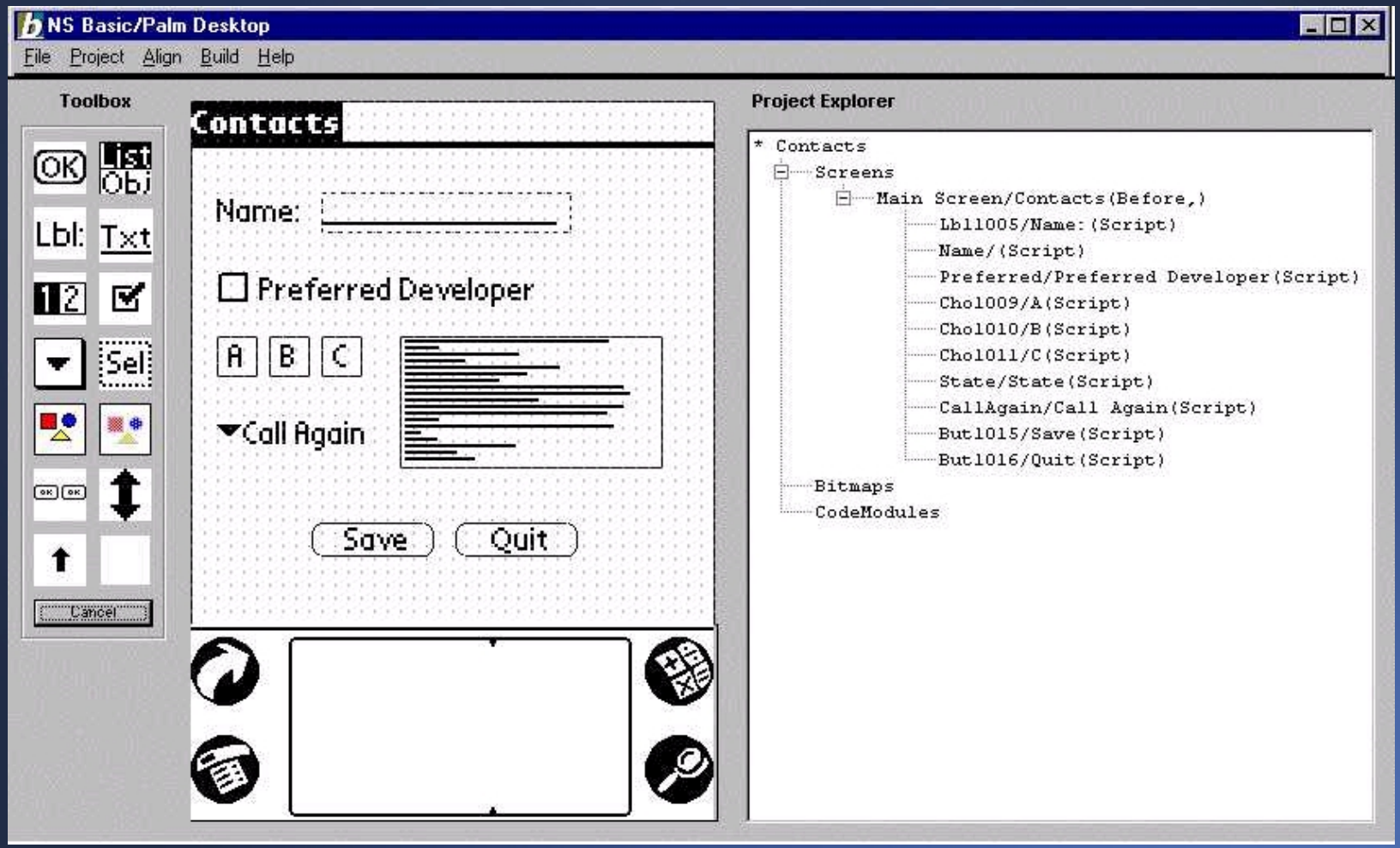
NS Basic

- Visual Basic Style IDE for Palm OS®
- Over 150 functions and statements
- Create simple application in an hour or less
- Cost $99.95

NS Basic

- Features
  - Standard and structured BASIC support
  - Over a dozen screen objects
  - Create buttons, fields and more
  - Produces standalone Palm apps
  - Serial Communication support
  - Math and trigonometric function support
  - Built-in database support

NS Basic


NS Basic/Palm Desktop
File Project Align Build Help
Toolbox
OK
List Obj
Lbl:
Txt
1 2
Sel
Cancel
Contacts
Name:
Preferred Developer
A B C
Call Again
Save
Quit
Project Explorer
* Contacts
Screens
Main Screen/Contacts(Before,)
Lbl1005/Name:(Script)
Name/(Script)
Preferred/Preferred Developer(Script)
Chol009/A(Script)
Chol010/B(Script)
Chol011/C(Script)
State/State(Script)
CallAgain/Call Again(Script)
But1015/Save(Script)
But1016/Quit(Script)
Bitmaps
CodeModules

NS Basic

- Why you want to use NS Basic
  - Quick learning curve
  - Similar to Visual Basic
  - Little programming

- http://www.nsbasic.com

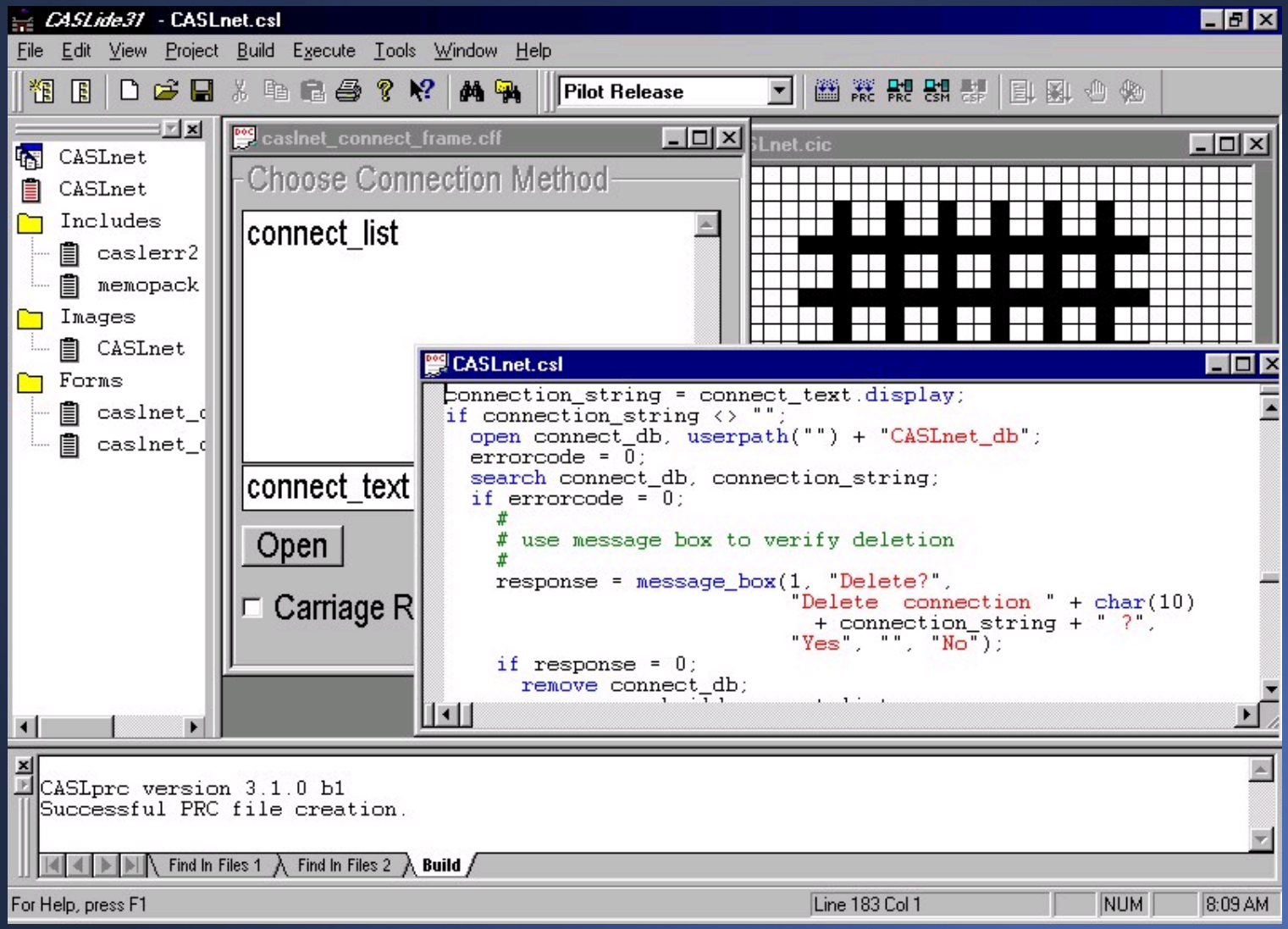
CASL

- Compact Application Solution Language: CASL
- Visual development tool
- Syntax is similar to IZL
- With or without runtime support
- Cost $64.95

CASL

- Features
  - Multi-Platform—Palm OS® , Windows, WinCE
  - Drag-n-Drop forms based development for UI design
  - Built-in database capabilities with a built-in conduit
  - Supports SMTP email, TCP/UDP network communications and serial communications
  - CASL runtime or standard C support (CASL Pro)
  - Freely available libraries to support printing (SCS Palmprinter or Symbol SPT devices)

CASL



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CASL

- Why you want to use CASL
  - Little programming
  - Fast with C code generation (Only in CASL Pro)
  - Libraries for printing via IR

- http://www.caslsoft.com

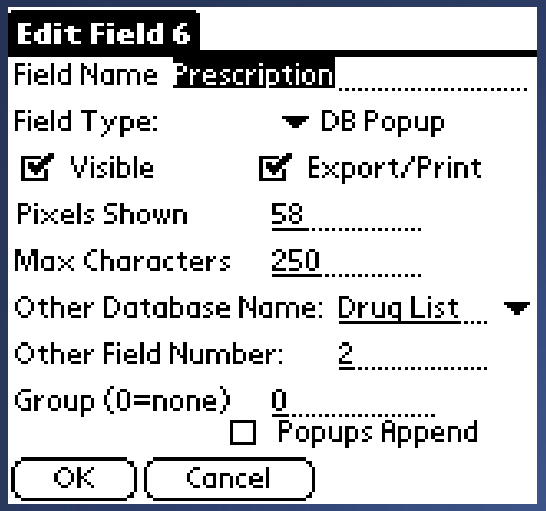
HanDBase

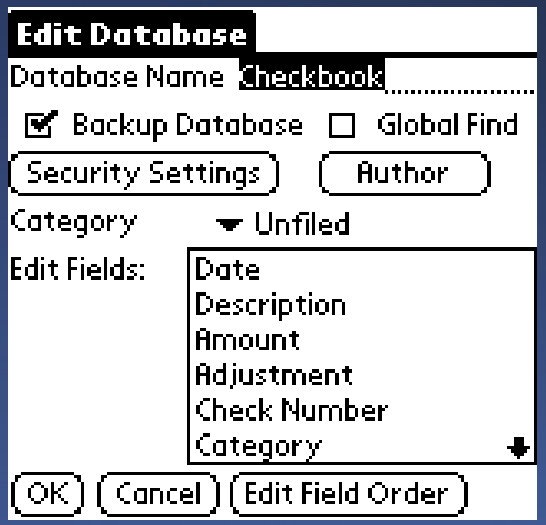
- Create, view and modify databases on your Palm OS®
- No coding necessary
- Development on the device
- Cost $24.99

HanDBase

- Features
  - Linked multiple relational database
    - Support up to 60 databases with 30 fields and can contain 65,000 records per database
  - Pop-up field, filtering, sorting
  - Exporting—Excel, CSW, Word, HTML, XML
  - Plug-In Architecture for third-party add-ons
  - Over 350 applets are available

HanDBase







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HanDBase

- Why you want to use HanDBase
  - Database forms
  - Flexibility to create, edit, and view databases in the field
  - Lots of fields (Date, Time, Pop-up, etc.)
  - Low cost $24.99

- http://www.ddhsoftware.com

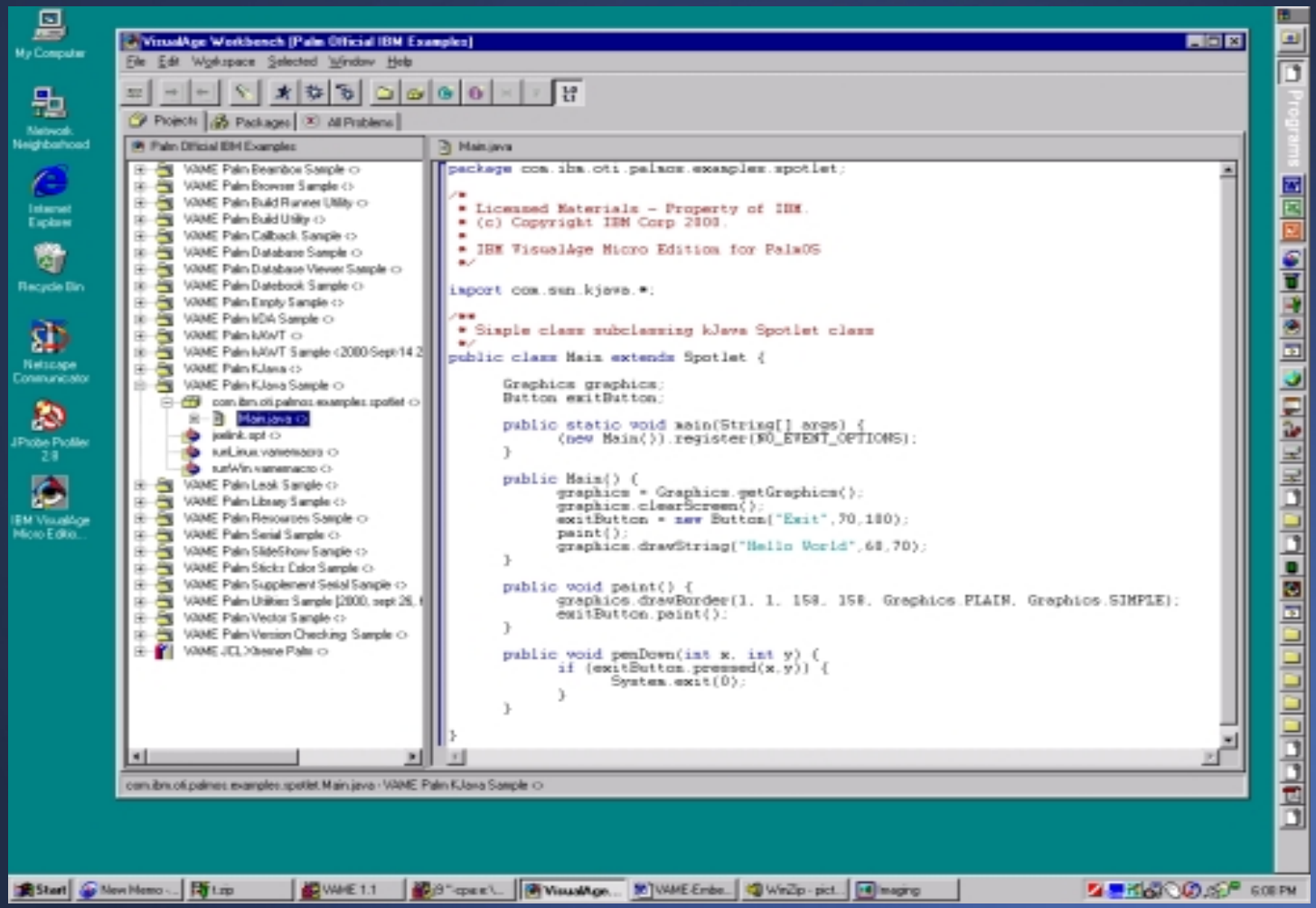
VisualAge Micro Edition for Palm OS

- Java development tools for Palm OS®
- Full Java support
- Support JDK 1.2.2
- Support Windows (Windows 2000) and Linux
- Cost around $99

VisualAge Micro Edition for Palm OS®

- Features
  - Java Native Interface (JNI) support
  - Garbage collection
  - Server support via TCP/IP
  - Optional dynamic class loading from class or jar files
  - Additional class library for Embedded Systems
  - Tools for code configuration, optimization and reduction tasks
  - IBM's VisualAge IDE

VisualAge Micro Edition for Palm OS



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VisualAge Micro Edition for Palm OS

- Why you want to use VisualAge
  - Java development
  - IBM support

- http://www.embedded.oti.com



Conduit Development

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Conduit Development Tools

- EHAND Connect
- Palm CDK
- Palm HotSync® Server



EHAND Connect

- Write custom conduits using Visual Basic or any COM-enabled language (Java,Delphi…)
- Easy to use, create a running conduit with less than 30 lines of code
- Free developer license with 2 restrictions
- Other licenses ($10 or $20 per end user)



EHAND Connect

- Features
  - Use Visual Basic to create conduits
  - Hides low-level complexity (byte ordering, calling conventions, etc.)
  - Easy integration with other Windows components via COM (ODBC, MS Access, MS Office, etc.)
  - Wizard conduit installation process
  - Includes full sample codes, thorough reference manual, and developer's guide



EHAND connect

- Why you want to use EHAND Connect
  - Fast custom conduit development
  - Easy to program (Visual Basic or COM-enabled)

- http://www.ehand.com



Palm CDK

- Palm Conduit Development Kit
  - C/C++ Sync Suite
    - C/C++ conduits, Mac and Windows
  - Sync Suite Java
    - Java conduits, Windows
  - COM Sync Suite
    - Visual Basic Conduits, Windows
- Free download
  - http://www.palmos.com



Palm CDK

- Features
  - Conduit Wizard and Starter Code
  - Sample code for all standard Palm OS® applications
  - SyncManager API
  - Synchronization Base Classes or Objects
  - Conduit Inspector for conduit debugging
  - Conduit Installer



Palm CDK

- Why you want to use the Palm CDK
  - Free
  - Sample Code, knowledge base, support
  - Visual C++, Visual Basic, Java

- http://www.palmos.com

Palm HotSync® Server

- Microsoft Exchange conduit (E-mail and calendar)
- Comprehensive console-based management
- Open architecture
- Wireline and Wireless support
- Available from selected Palm resellers
  - www.palm.com/products/enterprise/server.html

Palm HotSync® Server

- Features
  - Concurrently synchronize hundreds of Palm™ handhelds with enterprise data and services (email, calendars, databases, etc.)
  - Application development and customization
  - Centralized administration, backups, application distribution, etc.
  - TCP/IP support for remote synchronization

Palm HotSync® Server

- Why you want to use the Palm HotSync Server
  - Robust enterprise solution
  - Application development and customization

- www.palm.com/products/enterprise/server.html



Web Clipping Tools

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Web Clipping Tools

- Web Clipping Application Builder (WCA)
- Image Checker

WCA Builder

- Build WCA from HTML
- Simple and easy to use
- Rich set of tools to help authoring WCA
- Free from Palm, Inc.

WCA Builder

- Features
  - Converts HTML (3.2) to WCA
  - Support tables (no embedded tables)
  - Support color
    - Version 1.0.1—1, 2 bit
    - Version 1.5—1, 2, 4, 8, and 16 bit
  - Embed version number in the WCA
  - Support for Shift_Jis (Japanies character set, v1.5 only)
  - Data compression
    - Version 1.0.1—5 bit
    - Version 1.5—Lz77 (v2.0 of WCAs)

Image Checker

- Image preview for the WCA
- Let you see how it will look on the device
- Support GIF, JPEG, and BMP files
- Integrated in the Mac WCA builder

- http://www.palmos.com/dev/tech/webclipping


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